TEAM BIJAK MINDS PRESENTS

Orbyte Ecosystem ID: #416

KAK NADIAH ORBYTE SMART SME COPILOT

Autonomous Multimodal Copilot for MSMEs: Fusing Gemini Vision Intelligence, LHDN Mandatory e-Invoicing Compliance, and Resilient Multi-Dialect Governance.

COMPETING TEAM
BIJAK MINDS

TEAM LEAD & ARCHITECT
Ammar Hakim Bin Dzulhaidi

TARGET MARKET
97.4% of Malaysian MSMEs

THE CORE PROBLEM

MSME BOTTLENECKS: AUTOMATION BLINDSPOTS & COMPLIANCE

LHDN e-Invoicing Burden

With mandatory nationwide e-invoicing rollouts, traditional micro-merchants struggle with crumpled receipts, manual reconciliation, and complex tax formats.

82% MSMEs lack digital tax readiness

High Cost of Financial Talent

Hiring certified in-house accountants (avg. RM3,500+/mo) is unsustainable for hawkers, automotive workshops, and micro-retailers.

Audit compliance costs up 40%

The Language & Usability Barrier

Existing corporate SaaS solutions are English-only, rigid, and fail to comprehend local colloquial slang, Bahasa Melayu dialects, and mixed language usage.

SaaS adoption rate < 23% among MSMEs

INTEGRATED SOLUTION

KAK NADIAH: YOUR POCKET CHIEF OPERATING OFFICER

Beyond a Standard Chatbot

Kak Nadiah acts as an autonomous multimodal execution agent. She does not just chat; she perceives physical financial documents, cross-verifies ledger errors, and dispatches official business actions.

- **Multimodal Vision:** Converts crumpled invoices into compliant JSON within 3 seconds.
- **Localized Voice Interaction:** High-empathy native audio assistance via ElevenLabs.
- **Instant Automation:** Generates official vendor claims and emails documents automatically.

Synchronized with Orbyte Cloud (#416)

Seamlessly connected to the enterprise Orbyte Cloud ecosystem, ensuring zero data loss and bank-grade ledger traceability.

249+ Languages Supported	**100%** 2FA Audit Traceability
< 3s Vision Parse Speed	**RM0** Extra Hardware Cost

TECHNICAL ARCHITECTURE

CENTRALIZED TECH STACK & PRODUCTION PIPELINE

FRONTEND

Neo-Brutalism UI

HTML5, Tailwind CSS, Vanilla JS, Chart.js & Native Multilingual Translation Engine.

BACKEND

Node.js & Express

High-throughput REST API, secure ephemeral session parsing, and RFC 6238 TOTP protocols.

AI ENGINES

Gemini + ElevenLabs

Google Gemini 2.5 Flash (Vision/ OCR Reasoning) & ElevenLabs Multilingual v2 (Neural TTS).

GOVERNANCE

OTPLib 2FA + Resend

Google Authenticator verified ledger commits & transactional cloud email dispatch.

```
// DATA FLOW WORKFLOW
[Physical Receipt / Audio] ➔ [Gemini 2.5 Flash Vision Parser] ➔ [Structured e-Invoice JSON] ➔ [2FA Human-in-the-Loop] ➔ [Orbyte Cloud Ledger #416 + Resend API]
```

GOVERNANCE & SECURITY

HUMAN-IN-THE-LOOP: ZERO TAX PENALTY MOAT

⚠️ Risk of Unsupervised AI Hallucination

Standard generative AI models hallucinate numerical figures and date stamps. For SME accounting, false numbers lead to catastrophic penalties under tax audits.

Kak Nadiah strictly prohibits autonomous ledger commits without verification.

🛡️ Kak Nadiah Safeguard Protocol

1. **Discrepancy Detection:** Gemini flags mismatched totals, duplicate vendor names, or conflicting date lines.

2. **Mandatory 2FA Enforcement:** Privileged ledger commits strictly require time-based OTP via Google Authenticator.

3. **Immutable Audit Trails:** Cryptographic logging guarantees compliance transparency during tax checks.

DOMAIN 2 MUST-HAVE

THE 5-STEP CORE AUTOMATED JOURNEY

STEP 1

Trigger

Supplier invoice or raw receipt uploaded via web portal or WhatsApp scan.

Input Trigger

STEP 2

Data Ingestion

Gemini 2.5 Flash Vision extracts vendor, line-items, tax IDs, and billed units.

Structured JSON

STEP 3

Decision Logic

Cross-references contract benchmark rates & flags unauthorized markups.

Discrepancy Check

STEP 4

Human Approval

Owner reviews side-by-side card & authorizes with Google Authenticator (TOTP 2FA).

Human-in-the-Loop

STEP 5

Visible Outcome

Official claim letter dispatched via Resend API & permanently logged in Orbyte Cloud (#416).

Audit Trail Complete

ORBYTE ECOSYSTEM FIT Modular REST microservice built for zero-friction integration into shared SME business hubs.

100% Meets Domain 2 Rubric

FUTURE HORIZON & SDG

STRATEGIC MILESTONES & SCALABILITY (2026 – 2027)

Q4 2026

LHDN MyInvois API Gateway

Direct XML/JSON transmission to national tax servers with auto-validation status.

Q1 2027

WhatsApp Business Agent

Snap and send receipts directly on WhatsApp; audited into Orbyte Cloud instantly.

Q2 2027

Regional ASEAN Rollout

Expanding to Indonesia & Thailand via built-in 249+ language native translation.

UN SUSTAINABLE DEVELOPMENT GOALS (SDG 8 & SDG 9)

Democratizing enterprise-grade AI to bridge Malaysia's micro-enterprise productivity gap.

CLOSING PITCH

EMPOWERING MSMES, SECURING THE FUTURE!

"Not just an AI model, but a trusted and accountable co-pilot for every hardworking entrepreneur."

PROJECT RECAP:

Kak Nadiah – Orbyte Smart SME Copilot

SDC Hackathon 2026 • Xiamen University Malaysia

Team: Bijak Minds

Ready for Q&A